Central Bank of Armenia

SILENT RUNS IN A DOLLARIZED BANKING SYSTEM: DEPOSITOR-LEVEL EVIDENCE FROM FINANCIAL AND GEOPOLITICAL SHOCKS

WORKING PAPER WP-2025-06

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ABSTRACT

This paper examines depositor behavior during two episodes of system-wide withdrawal pressure in Armenia, a small, open, and highly dollarized economy. Using depositor-level administrative data covering 60–73 percent of the national deposit portfolio, we study early withdrawals during the 2014 exchange-rate and financial-market shock and the 2020 geopolitical shock, both characterized by sizable outflows despite the absence of bank-specific solvency stress. This environment allows us to analyze how contract characteristics, currency denomination, and depositor–bank relationships shape withdrawal decisions under uncertainty. We exploit Armenia's dual-currency deposit insurance regime to assess how currency-specific coverage interacts with exchange-rate risk, and we use legally defined insider classifications to identify individuals with formal ties to bank governance. We find that deposit insurance has a clear stabilizing effect, though insured depositors still adjust behavior modestly as balances approach the coverage limit. Insiders, despite lacking insurance and plausibly possessing superior information, do not withdraw more aggressively than the broader public. Prior crisis experience reduces the likelihood of withdrawal in subsequent shocks, and dual-currency depositors disproportionately redeem AMD deposits, consistent with depreciation-driven precautionary motives. These results underscore how currency composition, contract design, and relational ties shape withdrawal pressures under macro-financial stress.

Keywords:	Depositor Behavior, Deposit Insurance, Early Withdrawals, Financial Shock, Geopolitical Shock
JEL Codes:	D81, G01, G21

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1. INTRODUCTION

Episodes of sudden and sizable deposit withdrawals pose a longstanding challenge for financial stability. Whether concentrated at a single institution or unfolding as systemwide withdrawal waves, such events can strain bank liquidity, disrupt maturity transformation, and undermine public confidence. Historically, these vulnerabilities have shaped key regulatory reforms: early twentieth-century U.S. deposit insurance experiments sought to contain contagion but failed due to poor design, while the creation of the Federal Deposit Insurance Corporation (FDIC) in 1933 formalized a modern approach to safeguarding deposits in the aftermath of widespread bank failures.

Yet even with major advances in regulation and crisis management, recent episodes show that withdrawal pressures remain a major risk. During the 2008 global financial crisis, several institutions encountered acute liquidity stress amid abrupt outflows, and the 2023 collapse of Silicon Valley Bank demonstrated how large uninsured balances, rapid information flows, and digital banking technologies can amplify withdrawal dynamics. These modern episodes have renewed interest in the microeconomic drivers of depositor behavior and the conditions under which households and firms choose to withdraw funds.

Building on this renewed focus, the literature has long recognized that depositors may run not only in response to solvency concerns but also because of coordination failures or expectations about peer behavior, as emphasized in Diamond and Dybvig (1983). These mechanisms may become even more salient in emerging and highly dollarized economies, where exchange-rate volatility and external shocks can rapidly weaken confidence. Currency denomination adds another layer of heterogeneity: holders of foreign-currency deposits may react differently than domestic-currency depositors because of differing perceptions of inflation, depreciation risk, and institutional protection.

Understanding the determinants of early withdrawals is therefore essential for financial stability. Even in the absence of bank-specific distress, systemwide bank-run episodes triggered by macro shocks can force banks to liquidate assets and tighten liquidity conditions, generating broader spillovers. Deposit insurance schemes aim to mitigate such pressures, but their effectiveness depends on coverage limits, depositor sophistication, institutional design, and historical experience with instability. They may also introduce moral hazard by encouraging banks to take greater risks under the assumption that insured deposits are protected (Calomiris 1990). Their stabilizing role is further complicated in dollarized systems, where many depositors use foreign currency as a hedge against depreciation (Yeyati 2021). In such environments, withdrawal behavior is shaped not only by institutional protection but also by expectations about macroeconomic volatility and exchange-rate movements. Consistent with this, bank-level evidence shows that aggregate shocks can generate substantial deposit outflows even without idiosyncratic bank weakness (Levy-Yeyati, Martínez Pería, and Schmukler 2010), although depositor-level evidence remains limited.

The theoretical foundations for these phenomena are well established. Coordination failures, expectations, and information frictions figure prominently in models by Postlewaite and Vives (1987), Goldstein and Pauzner (2005), Calomiris and Kahn (1991), and Chen (1999). Empirical research using microdata shows that insured depositors may still withdraw preemptively (Iyer and Puri 2012; Iyer et al. 2016), that social interactions and information frictions can propagate panic (Starr and Yilmaz 2007; Kiss 2018), and that past crises shape trust and responsiveness (Fungáčová et al. 2019). Within this literature, depositor–bank relationships are often proxied through loan linkages, branch affiliation, or staff status, but such proxies do not identify the individuals who hold genuine decision-making roles within banks. Moreover, the literature rarely considers heterogeneity across currency denominations, a key omission for dollarized economies.

These theoretical and empirical insights take on renewed relevance considering recent bank failures. The Silicon Valley Bank episode underscored how uninsured balances and unrealized losses can magnify withdrawal risk (Choi et al. 2023), while other studies show that uninsured depositors respond more strongly to performance signals (Chen et al. 2022) and that declining franchise values and interest-rate exposure exacerbate liquidity pressures (Drechsler et al. 2023). Taken together, these findings demonstrate that depositor behavior and its interaction with institutional structure remain central to understanding the transmission of financial stress.

This paper contributes to this literature in several ways. First, rather than positioning the paper as an extension of classic bank-run theory, we anchor our analysis in the macroprudential and dollarization literatures, emphasizing how macro-financial shocks can generate systemwide withdrawal pressures even in the absence of bank-specific distress. In doing so, we provide one of the first depositor-level empirical characterizations of a “silent run” - a systemwide withdrawal wave without institution-specific solvency concerns that cannot be disentangled using bank-level data alone. Armenia, as a small, open, and highly dollarized economy structurally exposed to external disturbances, offers an informative setting for studying such behavior. The 2014 exchange rate and financial-sector turmoil and the 2020 geopolitical shock led to substantial deposit outflows across all banks, yet no institution exhibited solvency stress, allowing us to study depositor responses to aggregate uncertainty using micro-level data.

Second, we examine withdrawal behavior under a dual-currency deposit insurance scheme in which Armenian dram (AMD) and USD deposits face different coverage limits. This institutional feature provides a unique opportunity to study how currency-specific insurance interacts with exchange-rate risk - an empirical dimension largely absent from existing work.

Finally, we exploit highly granular administrative data that allow us to identify depositor–bank relationships with greater precision than in prior work. Building on staff-based classifications used in studies such as Iyer et al. (2016), our dataset includes a legally defined insider indicator that identifies depositors who are board members, major shareholders, senior managers, or their immediate family members. Although we do not separately observe each role, this classification captures the group of individuals with direct influence over bank decisions or privileged access to information. Since these insiders are not covered by deposit insurance, their behavior provides a unique perspective on how organizational proximity and informational advantage shape withdrawal decisions during periods of heightened uncertainty. This granular identification of insiders constitutes the paper's primary empirical contribution.

A distinctive feature of our empirical setting is that the two shocks we study differ fundamentally in their origins and transmission mechanisms. The 2014 episode was a financially driven exchange-rate and liquidity crisis rooted in external macroeconomic deterioration, Russian ruble depreciation, and dislocations in Armenia's foreign-exchange market. By contrast, the 2020 episode was not a financial panic but a geopolitical shock, in which precautionary withdrawals were motivated primarily by heightened uncertainty, security concerns, and broader disruptions to households' expectations.

The transmission of financial and geopolitical shocks, and the role of currency denomination and depositor–bank relationships, are especially salient given Armenia's institutional and historical context. Following the collapse of the Soviet financial system, the country experienced hyperinflation and a severe erosion of trust in banks. To rebuild confidence, Armenia established the Deposit Guarantee Fund in 2005, introducing differentiated insurance limits for AMD and foreign-currency deposits. Against this backdrop, the 2014 exchange-rate shock and the 2020 war shock triggered systemwide bank-run dynamics, offering a valuable opportunity to examine how depositors in a highly dollarized environment respond to different forms of macro-financial stress. Using administrative data covering approximately 60-73 percent of the country's deposit portfolio, we find that deposit insurance plays a stabilizing, though not fully comprehensive, role: insured deposits are noticeably more resilient than uninsured ones, yet withdrawal probabilities still increase, albeit modestly, as balances approach the coverage threshold. We show that despite being uninsured and arguably better-informed insiders do not systematically run ahead of the public. Their propensity to withdraw early is modestly higher than that of comparable outsiders during the 2014 shock and becomes indistinguishable in 2020, suggesting that close bank relationships and informational proximity have a stabilizing rather than destabilizing effect. In both shocks, larger deposit size, longer maturity, and exceeding the insurance threshold significantly increased the likelihood of early redemption, while higher interest rates reduced it. Importantly, we uncover evidence of learning effects: depositors who experienced the 2014 shock were less likely to withdraw early during the 2020 episode, suggesting a persistent behavioral adaptation rooted in past crisis experience. Moreover, dual-currency depositors, those holding both AMD and USD deposits, were more prone to withdrawing AMD deposits, consistent with precautionary motives amid local currency depreciation.

The paper is organized as follows. Section 2 introduces the institutional context in Armenia and describes the events under study. Section 3 outlines the data and methodology. Section 4 presents and discusses the results of the shocks. Section 5 introduces robustness checks, while Section 6 concludes.

2. BACKGROUND

2.1.Institutional Details

Armenia's banking system consists solely of private commercial banks. The primary regulatory body, the Central Bank of Armenia (CBA), maintains price and financial stability while overseeing prudential supervision, bank resolution, and the capital, liquidity, and foreign exchange exposures of banks. In line with Basel III standards, Armenian banks are subject to both the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR), which require institutions to hold sufficient high-quality liquid assets to withstand short-term liquidity stress and to maintain stable long-term funding structures. These requirements are particularly relevant in a dollarized financial system, where currency-specific liquidity pressures may emerge during periods of macro-financial stress.

Armenia's financial system is structurally characterized by a high level of dollarization. Households hold deposits and loans in both Armenian dram and foreign currencies, primarily USD, with euros used to a much lesser extent. During 2012–2020, approximately 65 percent of resident household deposits were denominated in foreign currency. Additionally, over 80 percent of all resident household deposits over the same period were time deposits, making them the most popular savings instrument. Depositors select the currency, principal amount, and term structure when opening accounts. Early withdrawals are permitted while maintaining full protection of principal; however, depending on the contract type, accrued interest may be forfeited. For example, most deposits forfeit all accumulated interest if withdrawn early, whereas contracts with regular interest payments (e.g., monthly or quarterly) only forfeit the interest for the current payment period.

The Deposit Guarantee Fund (DGF), an autonomous institution under CBA oversight established in 2005, administers Armenia's official deposit insurance scheme. Both private legal entities and individuals are covered, and the fund is financed through flat-rate premiums paid by commercial banks. Before November 2015, foreign-currency deposits were insured up to 2 million AMD, while dram deposits were covered up to 4 million AMD. When a depositor held both foreign-currency and dram accounts, compensation followed a two-tier logic: if the dram balance was below 2 million AMD, the foreign-currency balance was insured up to the difference between 2 million AMD and the compensated dram amount; if the dram balance exceeded 2 million AMD, only the dram portion was insured up to the 4 million AMD limit. Coverage increased to 10 million AMD for dram deposits and 5 million AMD for foreign-currency deposits in November 2015, and again to 16 million and 7 million AMD, respectively, in December 2020.

Although deposit insurance is legally guaranteed, payouts occur only after a bank is formally declared insolvent and claims are verified. During the resolution process, the CBA may temporarily restrict withdrawals while deciding whether to liquidate the institution, restructure it, or transfer its assets to a solvent bank. Because compensation is not instantaneous, even insured depositors may face periods of uncertainty, which can motivate precautionary early withdrawals despite formal coverage. To preserve depositor protection irrespective of borrowers' credit obligations, Armenian law prohibits banks from netting insured deposits against outstanding loans. Lending is not conditioned on deposit status, depositors do not acquire equity claims through their deposits, and there are no state-owned or cooperative commercial banks. Since expectations, perceived risk, and trust in the banking system drive variation in early withdrawal behavior, depositor characteristics do not systematically differ across banks.

2.2. Event Description

The shock that hit Armenia at the end of 2014 was multidimensional, transmitting through the real economy, external accounts, and the financial system simultaneously. Although Armenia is a net oil importer, the potential gains from lower energy prices were completely offset by the Russian economy recession. This was triggered by the combined effects of Western sanctions following the annexation of Crimea, the sharp decline in global oil prices, and a rapid loss of confidence in Russian financial markets. Russia being Armenia's dominant trading partner, investor, and source of remittances fell into recession, triggering a contractionary wave through trade, household incomes, and domestic demand. Exports to Russia declined by roughly a quarter, imports fell in parallel, and a drop in global metal prices reduced revenues from Armenia's key mining sectors. In effect, the external environment deteriorated across all fronts at once, creating a classic case of a small open economy exposed to highly correlated shocks.

Financial markets amplified these pressures. The rapid Russian ruble depreciation spilled over into Armenia's foreign exchange market, sharply increasing the demand for U.S. dollars amid heightened uncertainty. The dram depreciated significantly fueling speculation and eroding depositor confidence (see Figure 1). With a highly dollarized banking system, commercial banks became reluctant to sell dollars, a behavior that depositors interpreted as either precautionary or speculative hoarding. As a result, expectations of further depreciation intensified, queues formed at banks and exchange points, and the shortage of foreign currency became self-reinforcing.

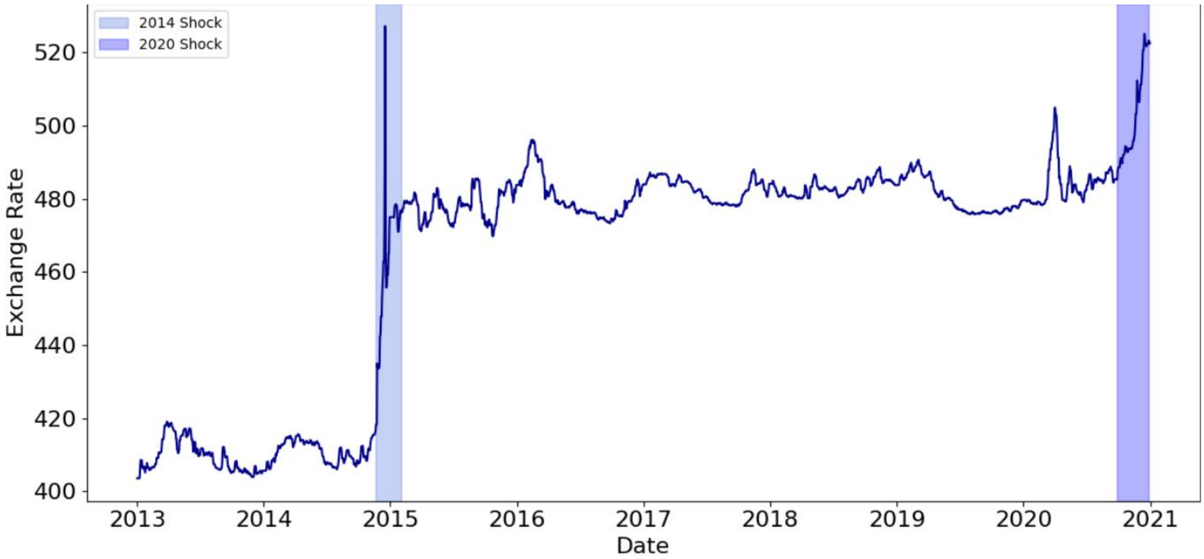
The situation escalated into a macroprudential challenge requiring decisive intervention. The CBA acted aggressively to stabilize expectations and contain liquidity stress. Policy rates were increased sharply, the key policy rate rose from 6.75% to 10.5% and the lombard rate from 8.25% to 21% reflecting a rapid repricing of risk as inflation expectations became unanchored. The CBA also conducted daily FX auctions and tightened reserve requirements to curb speculative positions. Crucially, foreign-currency reserve requirements were denominated in Armenian dram, meaning banks needed large amounts of local currency to meet their obligations. When the required reserve ratio on FX deposits doubled from 12% to 24%, banks faced significant dram liquidity needs, forcing them to enter the market to acquire drams amid depreciation pressures.

These policy actions ultimately prevented a deeper financial disruption. Markets stabilized, depositor panic subsided, and the exchange rate gradually returned to more sustainable levels, allowing the CBA to begin reversing the emergency rate hikes. The 2014 episode was far more than a routine exchange-rate adjustment: it was a full-fledged macro-financial shock that stressed the dollarized banking system, generated widespread depositor anxiety, and exposed structural vulnerabilities in Armenia's financial architecture.

The second significant shock to the Armenian financial system happened in late 2020, but its nature was fundamentally different from the currency-driven volatility of 2014. In September 2020, military war between Armenia and Azerbaijan started, coinciding with the continued COVID-19 outbreak. This combined shock generated major political, economic, and public-health issues, unsettling national security, straining public budgets, and generating anxiety among individuals and enterprises. Unlike 2014, the stress did not come from financial mismanagement or speculative currency spirals; rather, it arose from a rapid deterioration of the security environment paired with pandemic-related economic disruptions.

Following the onset of the conflict, the Armenian dram began to decline against major currencies, reflecting heightened market uncertainty and the impact of the combined geopolitical and epidemic shocks (Figure 1). The 2020 episode thus contrasts drastically with 2014: while 2014 displayed financially motivated panic induced by exchange-rate and solvency concerns, the 2020 episode was characterized by fear-driven, precautionary behavior during a significant national security crisis.[1]

Figure 1. AMD/USD Exchange Rate



Source: Central Bank of Armenia.

Notes. The figure shows the exchange rate of the Armenian dram per 1 U.S. dollar (AMD/USD). Highlighted periods correspond to major exchange-rate shocks described in the text. The first highlighted period reflects the end-2014 exchange rate shock, while the second period captures the late-2020 depreciation, driven by the geopolitical tensions and the ongoing COVID-19 disruptions.

[1] We do not analyze the COVID-19 period as a separate shock for several reasons. First, the exchange-rate depreciation during the initial phase of the pandemic was both smaller in magnitude and shorter in duration than in the 2014 and 2020 episodes (see Figure 1). The FX market stabilized quickly, and there is little evidence of prolonged financial-market stress. Second, withdrawal behavior in early 2020 was likely influenced by liquidity needs arising from lockdowns, income loss, and pandemic-related uncertainty rather than concerns about bank stability or currency devaluation. Because these mechanisms differ fundamentally from the macro-financial channels that motivate our analysis, we do not treat the COVID-19 depreciation as a separate shock episode.

3. DATA AND METHODOLOGY

3.1. Data

We analyze depositor withdrawal behavior using detailed depositor-level administrative data covering the period from 2012 through the end of 2020. The raw dataset is not a panel in the conventional sense; rather, it consists of the full contractual information for each term deposit - opening date, maturity date, currency, contracted interest rate, withdrawal date (if any), and the legally defined insider status of the depositor. The data further include key demographic characteristics, age, gender, and region (Yerevan, other regions of Armenia, and abroad), which we incorporate as controls throughout the analysis. For seven of the eight banks in 2014 and seven of the nine banks in 2020, we also observe within-contract balance adjustments, which allows us to reconstruct the evolution of deposit amounts over time. For the remaining banks, where intermediate balance updates are not available, we use the contracted opening balance to construct a simplified account history.[2] Using these transaction-level records, we create a panel-style sequence of each depositor's balances and inflow/outflow activity over the year leading up to each shock (i.e., from January 1, 2014 for the 2014 episode and from January 1, 2020 for the 2020 episode).

For empirical analysis, however, we do not estimate panel regressions. Instead, for each shock episode we build a separate cross-sectional dataset in which the unit of observation is a depositor–currency pair active at the beginning of the shock window. The reconstructed histories are used solely to compute pre-shock variables, such as the depositor's balance immediately before the episode, the interest rate in force, and measures of inflow and outflow activity earlier in the year. These variables summarize depositor behavior and contract conditions prior to the shock and allow us to examine how pre-shock characteristics shape early withdrawal decisions during the episode.

For the 2014 financial shock, we define the stress period as November 18, 2014 to February 10, 2015.[3] The corresponding cross-sectional dataset includes all deposits present during this interval from eight commercial banks, representing around 60 percent of total system

[2] For robustness, we re-estimate our main specifications in Appendix D, excluding the one bank in 2014 and the two banks in 2020 for which within contract balance adjustments are unavailable. The results are essentially unchanged, indicating that our findings are not driven by these institutions.

[3] The shock window ends on February 10, 2015, the date of the Central Bank of Armenia's Board announcement stating that the pressures in Armenia's financial and commodity markets stemming from late-2014 developments and the regional geopolitical situation were "gradually phasing out." See Central Bank of Armenia, Board Press Release, February 10, 2015, available at: https://www.cba.am/en/press-releases/1405

deposits and covering 51,750 unique depositors. Most depositors (95 percent) held deposits in only one currency.

For the 2020 geopolitical shock, we define the stress period as September 27, 2020 to December 31, 2020. Although hostilities ended on November 9, we extend the window through late December because uncertainty remained elevated and the exchange rate experienced renewed depreciation during this period. The cross-sectional dataset for this episode includes all deposits active during this window from nine banks, covering about 73 percent of total deposits and 73,661 unique depositors. Like 2014, 93 percent of depositors held single-currency accounts.

An important feature of our dataset is that it includes a bank-supplied indicator identifying depositors legally classified as insiders. These individuals hold positions or relationships that give them elevated informational proximity to the bank, and their deposits are not covered by deposit insurance.[4] This legal definition provides precise identification, in contrast with earlier studies that relied on proxies.[5] Insiders represent a very small share of depositors in both shocks accounting approximately 1.3 percent in 2014 and 1.6 percent in 2020.

The variable Above Insurance equals 1 if a depositor's total balance in a bank exceeds the official deposit insurance limit as of the event date. This distinction is important for testing whether insured and uninsured depositors respond differently to stress. Balances above the coverage threshold expose depositors to potential losses in case of bank distress, potentially making them more sensitive to shocks. We also include the Insured Balance variable, which measures the depositor's balance one week before the shock events. It equals the depositor's total balance if the account is below the insurance threshold, and zero otherwise.

[4] Armenian banking law provides a formal legal definition of *related parties*, set out in Article 8 of the Law on Banks and Banking. This category includes major shareholders (holding more than 20 percent of voting shares), board members, senior managers, certain employees who are deemed to influence core banking operations, and their immediate family members. These individuals are classified as related parties for regulatory purposes, and their deposits are not covered by deposit insurance. In our data, insider status is not constructed by the authors; rather, it is supplied directly by the participating banks, which identify related-party depositors according to this statutory definition. As a result, the insider indicator precisely captures those depositors who, by law, have governance influence or informational proximity to the bank.

[5] The literature uses a range of proxies to identify bank "insiders" or relationship depositors, including: borrowers who hold loans at the same bank (Iyer & Puri 2012, Iyer & Puri 2016), staff member (Iyer & Puri 2016), long-standing clients measured by tenure (Degryse & Ongena 2005). These proxies capture varying dimensions of relational proximity and informational advantage.

The variable Active Saver is a dummy equal to 1 for depositors who, over the year leading up to the shock, hold at least two deposit accounts that differ either by currency (AMD and USD) or by maturity structure (short-term and long-term), and 0 otherwise. This measure captures the depositor's level of financial engagement and diversification. Active savers diversifying across currencies or/and maturities are likely to manage their savings portfolio more strategically and may react differently to shocks than passive savers who maintain a single, homogeneous deposit.

The variable USD is an indicator equal to 1 if a depositor holds a USD-denominated deposit at the start of the shock period and 0 if the deposit is in Armenian drams. This variable accounts for potential behavioral differences between foreign- and domestic-currency depositors. Since expectations of depreciation, exchange rate volatility, and currency-specific risk perceptions may influence withdrawal decisions, distinguishing between AMD and USD deposits is important for capturing currency-driven heterogeneity in depositor responses.

Relationship Length measures the number of years between the shock year and the year when the depositor opened their first deposit account. It reflects the duration of the depositor's relationship with the banking system. Longer-standing clients may have developed trust and stability in their behavior, while newer depositors whose relationship with the bank is less established may exhibit more reactive or precautionary withdrawal patterns.

The variable Number of Accounts records the number of deposit accounts a depositor held historically prior to the shock. A larger number of accounts can indicate greater financial sophistication or segmentation of savings for different purposes.

The Interest Rate variable represents the weighted average interest rate across all deposits held by a depositor one week before the shock events. Because individuals may hold multiple accounts with different rates and maturities, this measure captures their overall return environment. It allows us to investigate whether depositors enjoying higher returns exhibit different sensitivities to shocks, consistent with the trade-off between yield and perceived liquidity or risk.

Days to Maturity is calculated as the logarithm of the weighted average number of days remaining until deposit maturity one week before the shock events. The variable captures liquidity constraints: deposits with longer remaining maturities are often easier to redeem early because a smaller share of the total interest has been accrued and thus forfeited upon early withdrawal, whereas deposits approaching maturity are more costly to terminate prematurely, making withdrawal less attractive.

To capture dynamic saving and withdrawal behavior prior to the shock, we construct two variables. Inflow History measures the ratio of the average positive changes in a depositor's balance to their total balance on the event date, calculated from the beginning of the shock year up to the shock date. Conversely, Outflow History measures the ratio of average negative balance changes to the depositor's total balance over the same period. This variable captures pre-shock withdrawals. Together, inflow and outflow offer a dynamic characterization of depositor behavior in the months leading up to the shock.

Table 1 reports summary statistics of key depositor characteristics for both shocks. In 2014, the average insured balance among AMD depositors is 880,000 AMD (approximately 1,830 USD), compared to around 2mln AMD (approximately 4160 USD) in the 2020 sample. For USD deposits, the average insured balance is 700 USD in 2014 and 2,600 USD in 2020.[6] The average depositor in 2020 is younger and has a longer relationship with the banking system, reflecting the continued expansion of formal financial participation over the decade. Deposit structures also differ markedly across shocks: depositors in 2020 hold a larger number of accounts on average and are more likely to be classified as active savers, consistent with a broader shift toward more diversified household financial portfolios. Contract characteristics show clear intertemporal variation as well: interest rates are substantially lower in 2020 for both AMD and USD deposits, maturity profiles are slightly shorter, while the share of uninsured deposits declines from 31 percent to 20 percent, partly due to the expansion of insurance limits in 2015. Early withdrawal rates also differ: they spike sharply during the 2014 exchange-rate shock but remain lower and less volatile during the 2020 geopolitical episode.

Despite the richness of the data, several limitations should be acknowledged. First, the dataset does not allow us to track the same individual across multiple banks, limiting our ability to observe complete financial behavior or cross-bank risk responses. This is a notable limitation because depositor reactions to stress may involve reallocating funds across banks rather than exiting the system altogether. Moreover, the inability to link individuals across institutions prevents us from observing currency-switching behavior that occurs through transfers between banks. Such adjustments are an important margin of depositor response during episodes of exchange-rate pressure. Also, without cross-bank identifiers, we cannot measure portfolio reshuffling such as splitting funds across several banks to reduce perceived risk.

Second, although the dataset includes broad regional indicators (Yerevan and regions), it lacks more granular geographic information such as neighborhood- or community-level

[6] USD equivalents are computed using an exchange rate of 1 USD = 480 AMD, and is used solely for comparability across currency denominations.

identifiers. This restricts our ability to study social network effects, local spillovers, or spatial heterogeneity in depositor behavior, which may be more nuanced than the available regional categories allow.

Table 1. Summary Statistics for 2014 and 2020 shocks

	2014		2020	
	Mean	SD	Mean	SD
Not Currency Specific				
Bank Insider	0.013	0.112	0.016	0.126
Male	0.438	0.496	0.408	0.492
Age	54.546	16.602	49.467	16.160
Relationship Length (in years)	1.950	1.933	3.984	3.556
Number of Accounts	4.018	6.153	6.593	11.627
Active Saver	0.379	0.485	0.342	0.475
Yerevan	0.503	0.500	0.547	0.498
Above Insurance	0.306	0.461	0.199	0.399
Currency: AMD				
Insured Balance (in hundred-thousand AMD)	8.752	11.179	19.683	24.670
Early Withdrawal	0.041	0.199	0.028	0.164
Interest Rate (%)	12.400	1.884	8.971	1.056
Days to Maturity (days in logarithms)	5.379	1.435	5.377	1.129
Inflow History	0.018	0.081	0.016	0.139
Outflow History	-0.024	0.450	-0.020	0.290
Currency: USD				
Insured Balance (in hundred-thousand USD)	0.007	0.012	0.026	0.033
Early Withdrawal	0.044	0.204	0.045	0.207
Interest Rate (%)	6.698	1.518	3.802	1.029
Days to Maturity (days in logarithms)	5.175	1.325	5.436	1.037
Inflow History	0.029	0.627	0.018	0.731
Outflow History	-0.032	0.651	-1.298	0.469

Source: Authors' calculations.

Note: This table reports summary statistics for depositor- and account-level variables used in the analysis for the 2014 and 2020 shock periods. Definitions of the variables can be found in Appendix A.

3.2.Methodology

To examine the determinants of early withdrawal during the shock periods, we estimate a linear probability model in which the dependent variable indicates whether a depositor withdraws a substantial share of their funds during the episode.[7] Specifically, the

[7] We estimate linear probability models because our primary interest lies in the sign and magnitude of the marginal effects, which are directly interpretable in this framework. In nonlinear models such as logit or probit, the marginal

dependent variable equals 1 if a depositor redeems at least 50 percent of their total balance during the shock, and 0 otherwise. Our key explanatory variables capture the insured portion of the deposit (Insured Balance), whether the balance exceeds the insurance limit (Above Insurance), and whether the depositor is classified as a legally defined insider. The specification further includes standard account-level controls such as the offered interest rate, remaining days to maturity, as well as person-level controls including relationship length, number of accounts, active saver status, and past account activity along with demographic characteristics. Where applicable, the model also incorporates bank fixed effects. Formally, we estimate:

$$\boldsymbol{Y_{bci} = \beta_0 + \beta_1 \text{InsuredBalance}_{ci} + \beta_2 \text{AboveInsurance}_{ci} + \beta_3 \text{Insider}_c + X_{bci} + D_c + \delta_b + \epsilon_{bci}}$$

Y_{bci} equals 1 if depositor i withdraws at least 50 percent of their total balance during the shock. X_{bci} includes the remaining depositor and account characteristics, D_c denotes demographic controls (age, gender, region), δ_b captures bank fixed effects when included and ϵ_{bci} is the error term.

In addition to the main analysis, we investigate how past experiences shape depositor behavior. Specifically, we investigate whether prior experience with a shock affects the likelihood and magnitude of early withdrawals, allowing us to assess how depositor behavior evolves in response to repeated banking stress events. We also explore the long-term influence of historical events, such as the Soviet-era hyperinflation, on current depositor decisions, testing whether prior exposure to high inflation affects risk perceptions and withdrawal behavior today.

4. RESULTS AND DISCUSSION

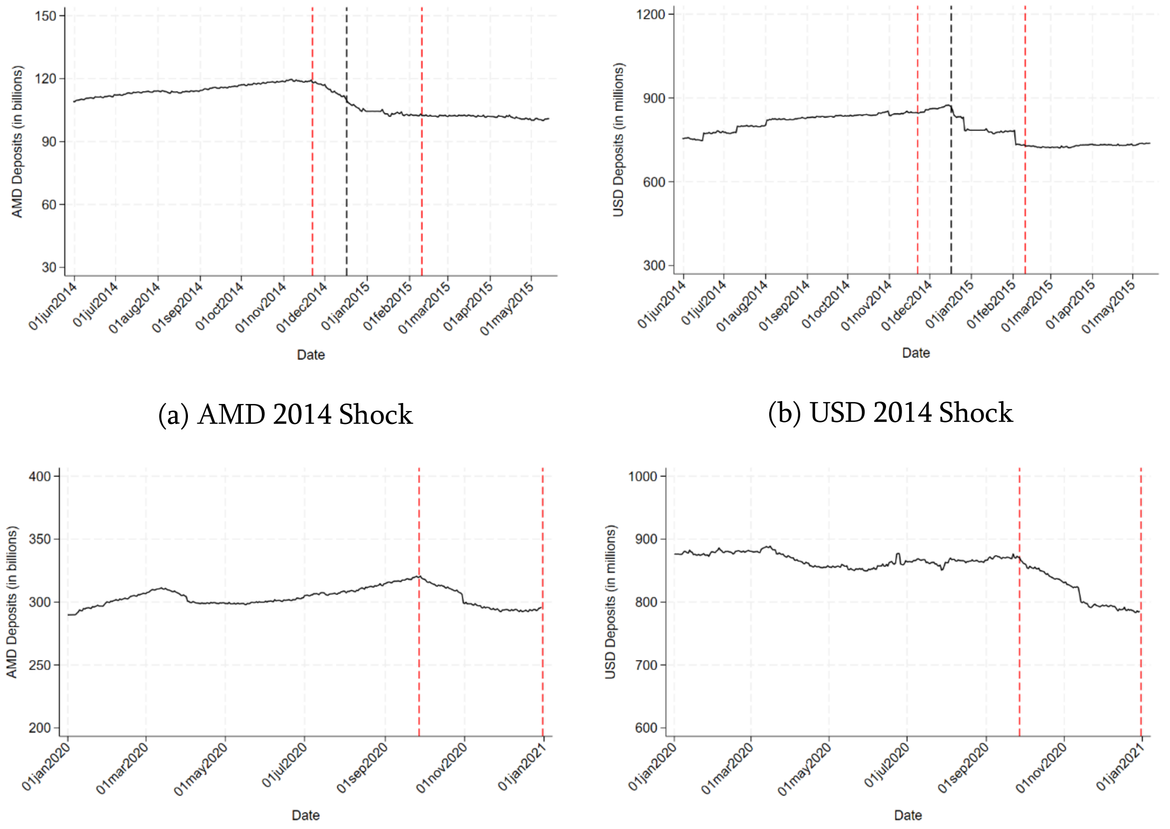
We begin by documenting the aggregate withdrawal patterns around the two shock episodes. Figure 2 presents the evolution of AMD and USD deposits across all banks during the 2014 and 2020 shocks, shown separately in four panels. In the 2014 episode, AMD deposits (panel a) began to decline starting around November 19 and fell sharply, decreasing by 13.7% by February 10, 2015. USD deposits (panel b) initially increased as

effects, particularly for interaction terms, depend on the full set of covariates and cannot be read directly from the estimated coefficients (Ai and Norton 2003; Norton, Wang, and Ai 2004), making interpretation considerably more cumbersome without altering the qualitative conclusions. As a robustness check, in Appendix C we also estimate logit models and report average marginal effects, which yield results that are very similar in both sign and significance.

depositors shifted into foreign currency, but after the major exchange-rate spike on December 17, they also began to drop, ultimately declining by 16.6% by February 10.

During the 2020 shock, deposit behavior was more gradual. AMD deposits (panel c) fell steadily from September 27 through December 30, with a total decrease of 7.4%. USD deposits (panel d) showed a similar pattern but a somewhat larger decline of 9.8% over the same period. Overall, deposit outflows in 2020 were smoother and less abrupt than those observed during the 2014 shock.

Figure 2. Deposit Amounts for All Banks by Currency During the 2014 and 2020 Shocks.



(a) AMD 2014 Shock

(b) USD 2014 Shock

(c) AMD 2020 Shock

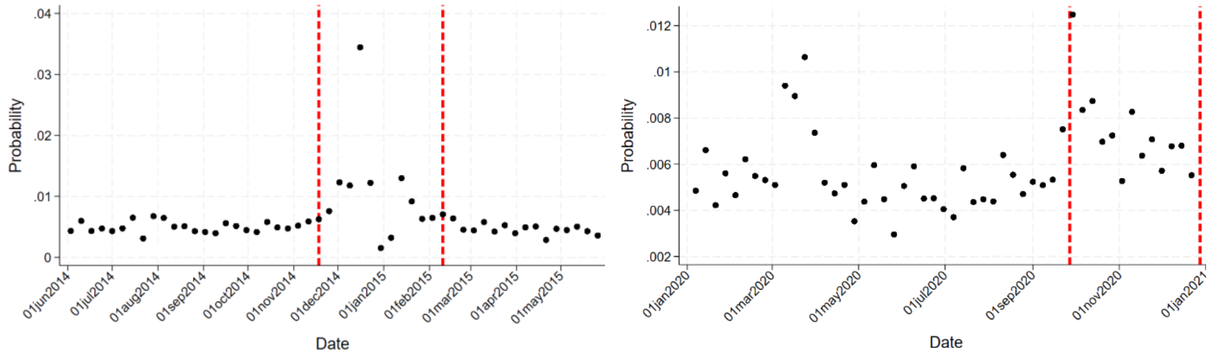
(d) USD 2020 Shock

Source: Authors' estimations.

Note: Axes are scaled and do not originate at zero.

As shown in Figure 3, the weekly probability that a depositor withdraws prematurely from at least one account rises sharply during episodes of macroeconomic disruption. Withdrawal rates remain stable prior to each shock and increase only once the respective crises begin, indicating that the events were effectively unanticipated at the household level. The 2014 exchange-rate shock generates a pronounced and immediate spike in early withdrawals, while the increase during the 2020 war shock is more moderate but still clearly timed with the onset of the conflict. Between these episodes, early-withdrawal rates revert to their usual low levels, suggesting that depositor behavior normalizes in the absence of major stress events.

Figure 3. Probability of Early Withdrawals during the 2014 and 2020 shocks.



(a) 2014 Shock (b) 2020 Shock

Source: Authors' estimations.

4.1. The 2014 Exchange-Rate Shock

The regression results reported in Table 2 reveal several systematic patterns in early-withdrawal decisions for both AMD and USD deposits.[8]

A key finding concerns the behavior of deposits below the insurance threshold. The variable Insured Balance enters positively and significantly, indicating that even within the insured range, larger deposits are more likely to be withdrawn. We interpret its coefficient separately for AMD and USD deposits to ensure meaningful comparisons because it is measured in units of 100,000. For AMD deposits, a 100,000 AMD increase in insured balance raises the probability of early withdrawal by approximately 0.1 percentage point. For USD deposits, we use an equivalent increase of 200 USD, roughly comparable in value to 100,000 AMD, which raises the probability of early withdrawal by approximately 0.16 percentage points. Although these marginal effects are modest, they indicate that insured depositors adjust their behavior systematically as their balances approach the limit of coverage.

Consistent with this interpretation, the Above Insurance indicator is strongly positive: uninsured deposits are substantially more likely to be withdrawn. After controlling bank fixed effects, the estimates in columns (3) and (6) indicate that uninsured AMD deposits are about 20 percentage points more likely to be withdrawn during the shock, whereas uninsured USD deposits are roughly 10.8 percentage points more likely to be redeemed.

[8] For readability, Table 2 includes only the key variables of interest; the full set of regression results is presented in Appendix Table B1.

This asymmetry suggests that USD depositors reacted less strongly, which is consistent with the nature of the episode: the sharp depreciation of the AMD made the shock relatively favorable for USD holders, whose balances appreciated in domestic-currency terms. Together, these results reveal a clear gradient of sensitivity to risk: fully insured deposits are the most stable, deposits approaching the coverage threshold begin to respond, and uninsured deposits, particularly in AMD, exhibit the highest propensity to withdraw.

Table 2. Regression Results for the 2014 Exchange Rate Shock

	AMD			USD		
	(1)	(2)	(3)	(4)	(5)	(6)
Insured Balance	0.0010***	0.0010***	0.0010***	0.7950***	0.7970***	0.8370***
	(0.0001)	(0.0001)	(0.0001)	(0.1680)	(0.1680)	(0.1710)
Insider	0.0320**	0.0316**	0.0280**	0.0257	0.0257	0.0241
	(0.0137)	(0.0137)	(0.0138)	(0.0163)	(0.0163)	(0.0162)
Above Insurance	0.0566***	0.0569***	0.0553***	0.0401***	0.0402***	0.0436***
	(0.0039)	(0.0039)	(0.0040)	(0.0046)	(0.0046)	(0.0048)
Inflow History		0.0356***	0.0339***		0.0008	0.0016
		(0.0129)	(0.0128)		(0.0166)	(0.0164)
Outflow History		0.0034	0.0034		−0.0003	0.0007
		(0.0022)	(0.0021)		(0.0169)	(0.0168)
Observations	37,856	37,856	37,856	16,756	16,756	16,756
R-squared	0.03	0.03	0.03	0.02	0.02	0.03
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	No	Yes	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2014 exchange rate shock, covering the period from November 18, 2014 to February 10, 2015. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Active Saver, Relationship Length, Number of Accounts, Interest Rate, and Days to Maturity. Columns (3) and (6) additionally include bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Turning to depositor categories, insiders do not exhibit strong or systematic early withdrawal behavior. Although these deposits belong to individuals with legally defined insider status and are explicitly uninsured, their coefficients are modest in magnitude and only marginally significant. This suggests that organizational proximity and privileged information do not lead insiders to run ahead of the public.

Table 3 further examines the behavior of USD depositors whose balances were close to the insurance threshold. The Barely variable identifies deposits just below the coverage limit (between 1,500,000 and 2,000,000 AMD, compared with the USD insurance ceiling of 2,000,000 AMD).

Table 3. Regression Results for 2014 Shock Including USD Depositors Just Below Insurance Coverage

	(1)	(2)	(3)
Barely	0.0295***	0.0292***	0.0309***
	(0.0071)	(0.0071)	(0.0072)
Above Insurance	0.0394***	0.0390***	0.0427***
	(0.0050)	(0.0049)	(0.0052)
Inflow History		0.0088	0.0094
		(0.0105)	(0.0104)
Outflow History		0.0085	0.0092
		(0.0107)	(0.0107)
Observations	13,477	13,477	13,477
R-squared	0.02	0.02	0.02
Additional Controls	Yes	Yes	Yes
Bank FE	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal of USD holders during the 2014 exchange rate shock, covering the period from November 18, 2014 to February 10, 2015. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. Barely is 1 if the savings account was 20% close to the insurance threshold from below and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Insured Balance, Active Saver, Relationship Length, Number of Accounts, Interest Rate, Days to Maturity, Insider status. Column (3) additionally includes bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The estimated coefficient on Barely is positive and significant, but notably smaller than the coefficient for Above Insurance. This pattern is consistent with the structure of incentives faced by these depositors. Because the local currency depreciated by roughly 20 percent during the shock, only a small subset of USD depositors would have recalculated their coverage in real time and recognized that their AMD-equivalent balances might temporarily exceed the insured limit. Most depositors are unlikely to perform such adjustments, meaning that "nearly insured" accounts behave more like insured deposits than like fully uninsured ones. Consequently, the withdrawal response among the Barely group is weaker than among depositors whose balances unambiguously exceeded the insurance ceiling.

Next, we examine whether withdrawal behavior differs for individuals with memories of hyperinflation. To isolate the effect of historical exposure from pure age differences, we control for age and age squared and include an indicator for belonging to the hyperinflation cohort - defined as individuals born in or before 1990, the period marking the collapse of the Soviet Union and the onset of severe monetary instability. The results in Table 4 indicate that members of this cohort exhibit a modestly higher likelihood of early withdrawal during shock periods. A plausible interpretation is that individuals who directly experienced, or grew up in households affected by, the hyperinflation episode may remain more sensitive to macroeconomic uncertainty and therefore more prone to precautionary responses. However, this evidence should not be viewed as causal: the hyperinflation variable may capture a combination of historical memory, household narratives, and unobserved demographic differences correlated with age. Rather, the findings provide suggestive evidence that exposure, whether direct or indirect, to the early-1990s macro-financial turmoil is associated with a slightly higher propensity to withdraw early during later episodes of stress.

Table 2. Early Withdrawal Behavior and Exposure to the 1990s Hyperinflation

	(1)	(2)	(3)
Hyperinflation Memory Group	0.0148***	0.0148***	0.0129**
	(0.0051)	(0.0051)	(0.0051)
Inflow History		0.0053	0.0051
		(0.0035)	(0.0034)
Outflow History		0.0029	0.0028
		(0.0020)	(0.0020)
Observations	54,612	54,612	54,612
R-squared	0.02	0.02	0.02
Additional Controls	Yes	Yes	Yes
Bank FE	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2014 exchange rate shock, covering the period from November 18, 2014 to February 10, 2015. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. Hyperinflation equals 1 for individuals who were born during or before the collapse of the Soviet Union. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Insured Balance, Above Insurance, Active Saver, Relationship Length, Number of Accounts, Interest Rate, Days to Maturity, and Insider status. Column (3) additionally includes bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.2.The 2020 Geopolitical Shock

Turning to the 2020 geopolitical shock, the nature of the event was fundamentally different from 2014. Rather than a conventional financial disturbance, the war shock represented an existential risk to the country. This distinction makes the 2020 episode particularly valuable for understanding how non-financial, high-stakes uncertainty shapes depositor behavior. Despite these differences, the patterns in Table 5 closely resemble those observed in 2014, suggesting that core behavioral mechanisms of withdrawal remain robust to the type of shock.[9]

Table 5. Regression Results for the 2020 Geopolitical Shock

	AMD			USD		
	(1)	(2)	(3)	(4)	(5)	(6)
Insured Balance	0.0002***	0.0002***	0.0002***	0.1150**	0.1140**	0.1890***
	(0.0000)	(0.0000)	(0.0000)	(0.0469)	(0.0469)	(0.0477)
Insider	0.0118*	0.0116*	0.0075	-0.0027	-0.0027	-0.0062
	(0.0069)	(0.0069)	(0.0070)	(0.0115)	(0.0115)	(0.0113)
Above Insurance	0.0234***	0.0235***	0.0267***	0.0345***	0.0345***	0.0393***
	(0.0028)	(0.0028)	(0.0028)	(0.0037)	(0.0037)	(0.0038)
Inflow History		0.0220*	0.0194*		-0.0001	-0.0001
		(0.0119)	(0.0103)		(0.0009)	(0.0006)
Outflow History		0.0071**	0.0064**		0.0000***	0.0000***
		(0.0031)	(0.0030)		(0.0000)	(0.0000)
Observations	56,348	56,348	56,348	22,743	22,743	22,743
R-squared	0.02	0.02	0.02	0.03	0.03	0.04
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	No	Yes	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2020 geopolitical shock, covering the period from September 27, 2020 to December 31, 2020. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Active Saver, Relationship Length, Number of Accounts, Interest Rate, and Days to Maturity. Columns (3) and (6) additionally include bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

[9] For readability, Table 5 includes only the key variables of interest; the full set of regression results is presented in Appendix Table B2.

Although the coefficients remain directionally consistent with those observed in 2014, their magnitudes are notably smaller. Two factors help explain this difference. First, as noted earlier, the 2020 geopolitical shock generated a substantially milder exchange rate movement compared to the pronounced depreciation during the 2014 financial shock. The weaker currency pressure in 2020 reduced the urgency for depositors to withdraw funds early. Second, our dataset covers deposit behavior only until the end of 2020, whereas the exchange rate remained depreciated roughly at the same level till May 2021. Consequently, part of the adjustment in depositor behavior likely occurred outside our observation window, leading to attenuated coefficient estimates relative to the 2014 episode.

Table 6 takes the analysis a step further by focusing on depositors who were active in both shocks. This allows us to ask whether past crisis exposure shapes future behavior. The evidence shows that depositors who experienced the 2014 shock are less likely to withdraw early during the 2020 war. This reduction in withdrawal probability indicates a behavioral adjustment: past crisis experience moderates the intensity of reactions to subsequent shocks.

Table 6. Early Withdrawal Behavior of Depositors with Prior Shock Experience

	(1)	(2)	(3)
Previous Shock Experience	-0.0093***	-0.0093***	-0.0092***
	(0.0020)	(0.0020)	(0.0021)
Inflow History		0.0015	0.0014
		(0.0019)	(0.0016)
Outflow History		0.0000***	0.0000***
		(0.0000)	(0.0000)
Observations	79,091	79,091	79,091
R-squared	0.02	0.02	0.03
Additional Controls	Yes	Yes	Yes
Bank FE	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2020 geopolitical shock, covering the period from September 27, 2020 to December 31, 2020. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. Previous Shock is 1 for individuals who had at least one saving account during the 2014 shock and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Insured Balance, Above Insurance, Active Saver, Relationship Length, Number of Accounts, Interest Rate, Days to Maturity, and Insider status. Column (3) additionally includes bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Two mechanisms may explain this pattern. The resilience mechanism suggests that depositors who withdrew during 2014 but did not experience losses updated their beliefs about the safety of the banking system. Having observed that early withdrawal was not necessary for loss avoidance, they may react less defensively in the future. The trust mechanism points to institutional credibility: the successful management of the 2014 shock by banks and the Central Bank may have strengthened depositor confidence. This accumulated trust reduces the likelihood of panic withdrawals in later episodes of instability.

As in 2014, no bank defaults occurred during the 2020 shock. Repeated exposure to crises without negative outcomes likely reinforced depositor confidence and contributed to a gradual buildup of resilience. Taken together, these results highlight that depositor behavior is path-dependent: past experiences shape how individuals respond to future shocks, even when those shocks differ fundamentally in nature and severity.

4.3. Two Currency Holders Across Two Shocks

Understanding differences between AMD and USD depositors is crucial, as the choice of deposit currency directly shapes risk perceptions, incentives, and behavioral responses during periods of stress. In a highly dollarized economy such as Armenia, USD deposits represent not only a financial instrument but also a form of self-insurance against exchange rate volatility. Consequently, USD and AMD depositors may react differently to shocks, with implications for the stability and liquidity of the banking system.

To deepen this analysis, we also study depositors who simultaneously hold accounts in both AMD and USD (Table 7). This group allows for a within-person comparison of how the same individual responds across currencies, shedding light on potential currency switching, portfolio reallocation, or asymmetric withdrawal behavior during shocks.

The nature of the shocks themselves played a key role in shaping these responses. For USD accounts, both the 2014 and 2020 shocks were effectively “positive” or wealth-enhancing shocks, as the USD appreciated against the AMD. During such episodes, USD depositors face weaker incentives to withdraw early; the appreciation increases the value of their deposits in AMD terms, providing an automatic buffer against uncertainty. In some cases, USD holders may even be inclined to maintain or augment their balances to continue benefiting from favorable exchange rate movements. The appreciation of their deposit currency functions as a built-in hedge, reducing panic-driven liquidation.

In contrast, AMD depositors are exposed to depreciation risk, which increases the opportunity cost of holding domestic-currency deposits during periods of instability. The depreciation of AMD erodes the real value of their savings, potentially prompting more defensive or precautionary behavior. For AMD accounts, the shocks represent a direct deterioration of the asset's purchasing power, making early withdrawal and subsequent conversion to foreign currency a rational response for many depositors.

These contrasting incentives become especially clear when examining depositors with dual-currency holdings. Because these individuals face identical personal circumstances and bank relationships across their accounts, differences in behavior can be attributed specifically to currency denomination. For these depositors, we observe heterogeneous responses: greater stability in USD accounts and more active reallocation or early withdrawal in AMD accounts during both shocks.

Table 7. Regression Results for Two-Currency Depositors: 2014 vs 2020 Shock

	2014			2020		
	(1)	(2)	(3)	(4)	(5)	(6)
USD	-0.1190***	-0.1170***	-0.1590***	-0.1020***	-0.0933***	-0.0917***
	(0.0161)	(0.0161)	(0.0210)	(0.0120)	(0.0126)	(0.0134)
Inflow History		0.2040***	0.1860**		0.2400**	0.2030**
		(0.0726)	(0.0725)		(0.1120)	(0.1010)
Outflow History		0.0489**	0.0440**		0.0000***	0.0000***
		(0.0211)	(0.0197)		(0.0000)	(0.0000)
Observations	5,791	5,791	5,791	10,874	10,874	10,874
R-squared	0.03	0.03	0.04	0.02	0.02	0.03
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	No	Yes	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2014 exchange rate and 2020 geopolitical shocks. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. USD is 1 for individuals who withdrew a USD account and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Insured Balance, Above Insurance, Active Saver, Relationship Length, Number of Accounts, Interest Rate, Days to Maturity, and Insider status. Columns (3) and (6) additionally includes bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Comparing the 2014 and 2020 episodes further highlights the importance of exchange rate magnitude. The 2014 financial shock involved a sharp depreciation of nearly 20 percent against the USD, creating significant pressure on AMD depositors and stronger incentives for early withdrawal. By contrast, the 2020 depreciation was milder (nearly 7 percent), reflecting a less severe currency impact. Moreover, as noted earlier, our dataset ends in

December 2020, while the depreciation and associated uncertainty continued into early 2021. This truncation may lead us to underestimate the full impact of the shock, since part of the behavioral response is not captured in our sample. As a result, the intensity of AMD withdrawal behavior was more pronounced in 2014, whereas the 2020 response, though still present, was more moderate.

Together, these findings show that currency denomination is a fundamental driver of depositor behavior. Exchange rate expectations, perceived safety of foreign versus domestic currency, and the interaction between depreciation risk and withdrawal incentives all play central roles in shaping depositor decisions during periods of elevated uncertainty.

5. ROBUSTNESS CHECKS

To ensure the robustness of our findings, we re-estimate our main specifications using several alternative definitions of early withdrawal. First, instead of classifying an early withdrawal as the premature liquidation of more than 50 percent of a depositor's balance, we lower the threshold to 25 percent. This broader definition captures a wider spectrum of precautionary or partial withdrawal behavior and allows us to assess whether our results are driven exclusively by large-scale liquidations or reflect more general depositor reactions to perceived risk.

Second, we introduce an alternative binary measure that equals 1 if a depositor fully withdraws at least one account, regardless of the withdrawn amount relative to the total portfolio. This specification identifies withdrawals even when only a small share of overall deposits is liquidated, thereby capturing extensive-margin responses that may be masked when focusing solely on aggregate balances.

Table 8 presents the results for the 2014 shock under these alternative definitions for both AMD and USD deposits. For each currency, the first column reports the baseline specification, while the second and third columns report the estimates using the 25 percent threshold and the "full withdrawal of at least one account" criterion, respectively. Across all specifications, the results remain highly stable: coefficient signs are unchanged, magnitudes are comparable, and key determinants of early withdrawal continue to be statistically significant. This indicates that our findings are not sensitive to the particular threshold used to define premature withdrawal. For the 2020 shock, we conduct the same robustness checks (Table 9). The pattern once again remains consistent: the direction and relative size of the key coefficients are stable across alternative definitions of early withdrawal.

Table 8. Alternative Outcome Variables for the 2014 Exchange Rate Shock

	AMD			USD		
	(1)	(2)	(3)	(4)	(5)	(6)
Insured Balance	0.00010***	0.00148***	0.00127***	0.837***	1.275***	1.471***
	(0.000109)	(0.000131)	(0.000181)	(0.171)	(0.200)	(0.320)
Insider	0.0280**	0.0480***	0.156***	0.0241	0.0266	0.0626**
	(0.0138)	(0.0161)	(0.0216)	(0.0162)	(0.0181)	(0.0245)
Above Insurance	0.0553***	0.0719***	0.0521***	0.0436***	0.0606***	0.0567***
	(0.00396)	(0.00461)	(0.00577)	(0.00481)	(0.00554)	(0.00868)
Inflow History	0.0339***	0.0512**	0.0821***	0.00157	0.00914	-0.0196
	(0.0128)	(0.0211)	(0.0290)	(0.0164)	(0.0165)	(0.0256)
Outflow History	0.00344	0.00522	0.00466***	0.000741	0.00847	-0.0195
	(0.00212)	(0.00318)	(0.00171)	(0.0168)	(0.0168)	(0.0256)
Observations	37,856	37,856	37,856	16,756	16,756	16,756
R-squared	0.04	0.04	0.04	0.03	0.03	0.03
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2014 exchange rate shock, covering the period from November 18, 2014 to February 10, 2015. In column (1) for each currency the dependent variable takes the value 1 if a depositor withdraws at least 50% of their total deposit balance during the shock period, and 0 otherwise. In column (2) for each currency, the dependent variable takes the value 1 if a depositor withdraws at least 25% of their total deposit balance during the shock period, and 0 otherwise. In column (3) for each currency the dependent variable takes the value 1 if a depositor fully withdraws at least one deposit account during the shock period, and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Active Saver, Relationship Length, Number of Accounts, Interest Rate, and Days to Maturity, as well as bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9. Alternative Outcome Variables for the 2020 Geopolitical Shock

	AMD			USD		
	(1)	(2)	(3)	(1)	(2)	(3)
Insured Balance	0.0002***	0.0004***	0.0005***	0.1890***	0.2930***	0.6970***
	(0.0000)	(0.0000)	(0.0001)	(0.0477)	(0.0524)	(0.0882)
Insider	0.00752	0.0105	0.0757***	-0.00615	-0.0124	-0.0269*
	(0.00698)	(0.00833)	(0.0124)	(0.0113)	(0.0126)	(0.0151)
Above Insurance	0.0267***	0.0439***	0.0299***	0.0393***	0.0579***	0.0689***
	(0.00283)	(0.00348)	(0.00434)	(0.00381)	(0.00419)	(0.00610)
Inflow History	0.0194*	0.0237**	0.0260	-0.0000636	0.0000403	-0.000493
	(0.0103)	(0.0116)	(0.0176)	(0.0006)	(0.0008)	(0.0009)
Outflow History	0.00640**	0.00920**	0.00430	0.0000***	0.0000***	0.0000***
	(0.00303)	(0.00404)	(0.00421)	(0.0000)	(0.0000)	(0.0000)
Observations	56,348	56,348	56,348	22,743	22,743	22,743
R-squared	0.02	0.03	0.06	0.04	0.05	0.06
Additional Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2020 geopolitical shock, covering the period from September 27, 2020 to December 31, 2020. In column (1) for each currency the dependent variable takes the value 1 if a depositor withdraws at least 50% of their total deposit balance during the shock period, and 0 otherwise. In column (2) for each currency, the dependent variable takes the value 1 if a depositor withdraws at least 25% of their total deposit balance during the shock period, and 0 otherwise. In column (3) for each currency the dependent variable takes the value 1 if a depositor fully withdraws at least one deposit account during the shock period, and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Active Saver, Relationship Length, Number of Accounts, Interest Rate, and Days to Maturity, as well as bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

In addition, we modify the end date of the shock period. Since on December 1, 2020, the deposit insurance threshold increased from 10 million AMD to 16 million AMD for AMD deposits, and from 5 million AMD to 7 million AMD for USD deposits, we test whether this policy change influences our results. In an alternative specification, we therefore restrict the shock period to end in late November, thereby excluding any effects associated with the revised insurance limits. The results reported in Table 10 demonstrate a high degree of consistency with our baseline findings.

We also estimate alternative logit models, presented in Appendix C, which similarly yield unchanged results. Finally, we re-run our baseline regressions excluding banks for which within-contract balance adjustments are unavailable; the findings again remain fully stable. Together, these exercises demonstrate that our results are robust across alternative outcome definitions, estimation methods, and sample restrictions.

Table 10. Alternative Shock Date for the 2020 Geopolitical Shock

	AMD		USD	
	(1)	(2)	(1)	(2)
Insured Balance	0.0002***	0.0002***	0.1890***	0.2020***
	(0.0000)	(0.0000)	(0.0477)	(0.0466)
Insider	0.00752	0.00602	-0.00615	-0.00407
	(0.00698)	(0.00692)	(0.0113)	(0.0115)
Above Insurance	0.0267***	0.0270***	0.0393***	0.0410***
	(0.00283)	(0.00284)	(0.00381)	(0.00374)
Inflow History	0.0194*	0.0198*	-0.0000636	-0.00002
	(0.0103)	(0.0104)	(0.0006)	(0.0006)
Outflow History	0.00640**	0.00670**	0.0000***	0.0000***
	(0.00303)	(0.00308)	(0.0000)	(0.0000)
Observations	56,348	56,348	22,743	22,743
R-squared	0.02	0.02	0.04	0.04
Additional Controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes

Source: Authors' estimations.
Note: This table reports the results of a regression examining the probability of a withdrawal during the 2020 geopolitical shock. In all columns the dependent variable takes the value 1 if a depositor withdraws at least 50% of their total deposit balance during the shock period, and 0 otherwise. In column (1) for each currency, the data covers the period from September 27, 2020 to December 31, 2020. In column (2) for each currency, the data covers the period from September 27, 2020 to November 30, 2020. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as all standard depositor and account characteristics, including Active Saver, Relationship Length, Number of Accounts, Interest Rate, and Days to Maturity, as well as bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.
Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

6. CONCLUSION

This paper provides new evidence on the microeconomic determinants of depositor behavior during episodes of heightened macro-financial uncertainty. Using granular administrative data covering most of the Armenian banking system, we examine two system-wide bank-run episodes - one driven by financial market distress in 2014 and the other by geopolitical uncertainty in 2020. Despite their different origins and transmission channels, both shocks generated sizable outflows in a highly dollarized environment without triggering institution-specific solvency problems. This unique setting allows us to isolate the behavioral drivers of early withdrawals in the absence of classic bank runs and to assess how contract characteristics, currency denomination, and depositor–bank relationships interact with macroeconomic uncertainty.

Several key insights emerge from our analysis. First, we find that deposit insurance reduces the likelihood of early withdrawal, but its stabilizing power is incomplete. Insured deposits are significantly more stable than uninsured ones, yet coverage limits do not eliminate withdrawal pressures during systemic shocks. These results underscore that deposit insurance remains an important stabilizing tool but cannot fully prevent withdrawals when economy-wide uncertainty rises.

Second, the information on depositor–bank relationships provides new insights into the behavior of insiders. Unlike most prior studies, which rely on staff status or indirect proxies for relational proximity, our data include a legally defined insider indicator that covers board members, major shareholders, senior managers, and certain key employees. Despite being uninsured and arguably better informed, insiders do not systematically run ahead of other depositors. Their withdrawal behavior is only modestly higher during the 2014 episode and becomes indistinguishable from that of outsiders during the 2020 shock. This pattern suggests that close organizational and informational ties may reduce, rather than amplify, panic-driven responses.

Third, historical experience plays a measurable role in shaping depositor responses to new shocks. Individuals who lived through the hyperinflation of the early 1990s exhibit a higher propensity to withdraw during the 2014 shock, suggesting that economic memory can heighten sensitivity to macroeconomic uncertainty. At the same time, we find evidence of learning across episodes: depositors who withdrew during the 2014 crisis were significantly less likely to do so during the 2020 shock.

Finally, dual-currency depositors further illustrate this mechanism: within the same individual, AMD deposits are disproportionately more likely to be withdrawn, reflecting precautionary motives rooted in exchange-rate expectations rather than concerns about bank solvency.

Taken together, our findings provide a richer understanding of depositor behavior under different forms of macroeconomic stress and shed light on the mechanisms through which currency risk, institutional design, and personal relationships interact to influence withdrawal decisions. In highly dollarized economies, where external shocks and exchange-rate volatility are recurring features of the macro-financial landscape, these insights are particularly relevant for the design of stability frameworks. Our results underscore the importance of liquidity regulation calibrated to currency-specific run risks: during short-lived but intense stress episodes, the LCR plays a critical role in ensuring that banks can absorb withdrawal pressures, and our evidence suggests that run-off rates may need to reflect currency denomination in dollarized systems. Although long-term funding rules such as the NSFR are less directly connected to short-run withdrawal dynamics, persistent patterns in depositor behavior, such as precautionary currency switching, may influence the composition and stability of funding over time. More broadly, the findings highlight the need for crisis-management and communication policies that recognize currency-specific trust, heterogeneous depositor sophistication, and the stabilizing role of insider relationships.

Overall, the study highlights that understanding depositor behavior requires moving beyond the question of whether runs occur to examine why, when, and under what conditions depositors choose to withdraw - insights that can meaningfully inform macroprudential policy in dollarized emerging economies.

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APPENDIX A

Variable	Definition
Above Insurance	A dummy equal to 1 if a depositor's total balance in a bank exceeds the official deposit insurance limit as of the event date, and 0 otherwise.
Insured Balance	The depositor's balance one week before the shock for accounts below the insurance limit (zero otherwise), expressed in hundred-thousand AMD for AMD deposits and hundred-thousand USD for USD deposits.
Active Saver	A dummy equal to 1 if, over the year preceding the shock, a depositor holds at least two term deposits that differ either by currency (AMD vs. USD) or by maturity structure (short-term vs. long-term), capturing pre-shock financial activeness and diversification.
Insider	Depositors legally classified as related parties to the bank, such as board members, major shareholders, senior managers, or their close family members, whose deposits are not covered by deposit insurance.
USD	A dummy equal to 1 if the depositor holds a USD-denominated deposit at the start of the shock period, and 0 for AMD-denominated deposits.
Relationship Length	The number of years between the shock year and the year in which the depositor opened their first deposit account, measuring the duration of the depositor-bank relationship.
Number of Accounts	The total number of deposit accounts the depositor has held historically in a bank prior to the shock, reflecting the extent of account diversification.
Interest Rate	The weighted average interest rate across all deposits held by the depositor one week before the shock period.
Days to Maturity	The logarithm of the weighted average number of days remaining until deposit maturity (plus one), measured one week before the shock.
Inflow History	The ratio of the depositor's average positive balance changes during the pre-shock year to their total balance at the event date; equals 0 if no inflows occur.
Outflow History	The ratio of the depositor's average negative balance changes during the pre-shock year to their total balance at the event date; equals 0 if no outflows occur.

APPENDIX B

Table B1. Full Regression Results for the 2014 Exchange Rate Shock

	AMD			USD		
	(1)	(2)	(3)	(4)	(5)	(6)
Insured Balance	0.0010***	0.0010***	0.0010***	0.7950***	0.7970***	0.8370***
	(0.0001)	(0.0001)	(0.0001)	(0.1680)	(0.1680)	(0.1710)
Active Saver	0.0300***	0.0294***	0.0285***	0.0227***	0.0227***	0.0234***
	(0.0025)	(0.0025)	(0.0025)	(0.0034)	(0.0034)	(0.0035)
Relationship Length	–0.0035***	–0.0034***	–0.0032***	0.0001	0.0001	–0.0004
	(0.0007)	(0.0007)	(0.0007)	(0.0009)	(0.0009)	(0.0009)
Number of Accounts	0.0014***	0.0014***	0.0014***	0.0007**	0.0007**	0.0004
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0002)
Interest Rate	–0.0051***	–0.0050***	–0.0058***	–0.0093***	–0.0093***	–0.0137***
	(0.0007)	(0.0007)	(0.0008)	(0.0011)	(0.0011)	(0.0019)
Days to Maturity	0.0071***	0.0072***	0.0063***	0.0101***	0.0102***	0.0111***
	(0.0006)	(0.0006)	(0.0006)	(0.0010)	(0.0010)	(0.0010)
Insider	0.0320**	0.0316**	0.0280**	0.0257	0.0257	0.0241
	(0.0137)	(0.0137)	(0.0138)	(0.0163)	(0.0163)	(0.0162)
Above Insurance	0.0566***	0.0569***	0.0553***	0.0401***	0.0402***	0.0436***
	(0.0039)	(0.0039)	(0.0040)	(0.0046)	(0.0046)	(0.0048)
Inflow History		0.0356***	0.0339***		0.0008	0.0016
		(0.0129)	(0.0128)		(0.0166)	(0.0164)
Outflow History		0.0034	0.0034		–0.0003	0.0007
		(0.0022)	(0.0021)		(0.0169)	(0.0168)
Observations	37,856	37,856	37,856	16,756	16,756	16,756
R-squared	0.03	0.03	0.03	0.02	0.02	0.03
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	No	Yes	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2014 exchange rate shock, covering the period from November 18, 2014 to February 10, 2015. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region. Columns (3) and (6) additionally include bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B2. Regression Results for the 2020 Geopolitical Shock

	AMD			USD		
	(1)	(2)	(3)	(4)	(5)	(6)
Insured Balance	0.0002***	0.0002***	0.0002***	0.1150**	0.1140**	0.1890***
	(0.0000)	(0.0000)	(0.0000)	(0.0469)	(0.0469)	(0.0477)
Active Saver	0.0182***	0.0181***	0.0188***	0.0263***	0.0263***	0.0266***
	(0.0017)	(0.0017)	(0.0018)	(0.0030)	(0.0030)	(0.0030)
Relationship Length	-0.0003	-0.0003	-0.0005**	-0.0010**	-0.0010**	-0.0009*
	(0.0002)	(0.0002)	(0.0002)	(0.0005)	(0.0005)	(0.0005)
Number of Accounts	0.0002**	0.0002**	0.0001	0.0004***	0.0004***	0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Interest Rate	-0.0133***	-0.0132***	-0.0147***	-0.0196***	-0.0196***	-0.0223***
	(0.0007)	(0.0007)	(0.0009)	(0.0014)	(0.0014)	(0.0016)
Days to Maturity	0.0090***	0.0090***	0.0109***	0.0135***	0.0135***	0.0167***
	(0.0005)	(0.0005)	(0.0005)	(0.0011)	(0.0011)	(0.0012)
Insider	0.0118*	0.0116*	0.0075	-0.0027	-0.0027	-0.0062
	(0.0069)	(0.0069)	(0.0070)	(0.0115)	(0.0115)	(0.0113)
Above Insurance	0.0234***	0.0235***	0.0267***	0.0345***	0.0345***	0.0393***
	(0.0028)	(0.0028)	(0.0028)	(0.0037)	(0.0037)	(0.0038)
Inflow History		0.0220*	0.0194*		-0.0001	-0.0001
		(0.0119)	(0.0103)		(0.0009)	(0.0006)
Outflow History		0.0071**	0.0064**		0.0000***	0.0000***
		(0.0031)	(0.0030)		(0.0000)	(0.0000)
Observations	56,348	56,348	56,348	22,743	22,743	22,743
R-squared	0.02	0.02	0.02	0.03	0.03	0.04
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	No	Yes	No	No	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2020 geopolitical shock, covering the period from September 27, 2020 to December 31, 2020. The dependent variable equals 1 if an individual withdraws more than 50% of their savings during the shock, and 0 otherwise. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region. Columns (3) and (6) additionally include bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

APPENDIX C

Table C1. Baseline vs. Logit Estimation for the 2014 Exchange Rate Shock

	AMD		USD	
	(1)	(2)	(1)	(2)
Insured Balance	0.0010***	0.00106***	0.8370***	0.9702***
	(0.0001)	(0.0001)	(0.1710)	(0.2202)
Active Saver	0.0285***	0.031***	0.0234***	0.0256***
	(0.0025)	(0.0025)	(0.0035)	(0.0037)
Relationship Length	–0.0032***	-0.00199***	–0.0004	-0.00036
	(0.0007)	(0.0007)	(0.0009)	(0.0080)
Number of Accounts	0.0014***	0.0004***	0.0004	0.00009
	(0.0003)	(0.0001)	(0.0002)	(0.0001)
Interest Rate	–0.0058***	-0.0051***	–0.0137***	-0.0125***
	(0.0008)	(0.0007)	(0.0019)	(0.0015)
Days to Maturity	0.0063***	0.0069***	0.0111***	0.0112***
	(0.0006)	(0.0007)	(0.0010)	(0.0009)
Insider	0.0280**	0.0092	0.0241	0.01460
	(0.0138)	(0.0067)	(0.0162)	(0.0089)
Above Insurance	0.0553***	0.04589***	0.0436***	0.0490***
	(0.0040)	(0.0032)	(0.0048)	(0.0066)
Inflow History	0.0339***	0.2892***	0.0016	0.001779
	(0.0128)	(0.0446)	(0.0164)	(0.0135)
Outflow History	0.0034	0.2696	0.0007	0.0013
	(0.0021)	(0.0471)	(0.0168)	(0.0137)
Observations	37,856	37,856	16,756	16,756
Demographic Controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2014 exchange rate shock, covering the period from November 18, 2014 to February 10, 2015. The dependent variable takes the value 1 if a depositor withdraws at least 50% of their total deposit balance during the shock period, and 0 otherwise. Column (1) for each currency reports estimates from a linear probability model. Column (2) for each currency reports the marginal effects from the Logit specification. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table C2. Baseline vs. Logit Estimation for the 2020 Geopolitical Shock

	AMD		USD	
	(1)	(2)	(1)	(2)
Insured Balance	0.0002***	0.0002***	0.1890***	0.1825***
	(0.0000)	(0.0000)	(0.0477)	(0.0650)
Active Saver	0.0188***	0.0179***	0.0266***	0.0277
	(0.0018)	(0.0016)	(0.0030)	(0.0032)
Relationship Length	-0.0005**	-0.0005**	-0.0009*	-0.0008*
	(0.0002)	(0.0002)	(0.0005)	(0.0004)
Number of Accounts	0.0001	0.0000	0.0001	0.0000
	(0.0001)	(0.0000)	(0.0001)	(0.0001)
Interest Rate	-0.0147***	-0.0145***	-0.0223***	-0.0257***
	(0.0009)	(0.0008)	(0.0016)	(0.0018)
Days to Maturity	0.0109***	0.0136***	0.0167***	0.0198***
	(0.0005)	(0.0007)	(0.0012)	(0.0015)
Insider	0.0075	0.0053	-0.0062	-0.0046
	(0.0070)	(0.0041)	(0.0113)	(0.0090)
Above Insurance	0.0267***	0.0219***	0.0393***	0.0367***
	(0.0028)	(0.0020)	(0.0038)	(0.0042)
Inflow History	0.0194*	0.0619	-0.0001	0.2540***
	(0.0103)	(0.0613)	(0.0006)	(0.0780)
Outflow History	0.0064**	0.0475	0.0000***	0.2297***
	(0.0030)	(0.0515)	(0.0000)	(0.0762)
Observations	56,348	56,318	22,743	22,705
Demographic Controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2020 geopolitical shock, covering the period from September 27, 2020 to December 31, 2020. The dependent variable takes the value 1 if a depositor withdraws at least 50% of their total deposit balance during the shock period, and 0 otherwise. Column (1) for each currency reports estimates from a linear probability model. Column (2) for each currency reports the marginal effects from the Logit specification. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

APPENDIX D

Table D1. Regression results for 2014 Exchange Rate Shock without the bank for which within-contract balance adjustments are unavailable

	AMD		USD	
	(1)	(2)	(1)	(2)
Insured Balance	0.0010***	0.0010***	0.8370***	0.8440***
	(0.0001)	(0.0001)	(0.1710)	(0.1710)
Active Saver	0.0285***	0.0285***	0.0234***	0.0235***
	(0.0025)	(0.0025)	(0.0035)	(0.0035)
Relationship Length	–0.0032***	-0.0034***	–0.0004	-0.0004
	(0.0007)	(0.0007)	(0.0009)	(0.0009)
Number of Accounts	0.0014***	0.0014***	0.0004	0.0003
	(0.0003)	(0.0003)	(0.0002)	(0.0002)
Interest Rate	–0.0058***	-0.0056***	–0.0137***	-0.0140***
	(0.0008)	(0.0008)	(0.0019)	(0.0020)
Days to Maturity	0.0063***	0.0064***	0.0111***	0.0115***
	(0.0006)	(0.0006)	(0.0010)	(0.0010)
Insider	0.0280**	0.0265*	0.0241	0.0249
	(0.0138)	(0.0137)	(0.0162)	(0.0164)
Above Insurance	0.0553***	0.0551***	0.0436***	0.0440***
	(0.0040)	(0.0040)	(0.0048)	(0.0048)
Inflow History	0.0339***	0.0337***	0.0016	0.0019
	(0.0128)	(0.0128)	(0.0164)	(0.0164)
Outflow History	0.0034	0.0034	0.0007	0.0011
	(0.0021)	(0.0021)	(0.0168)	(0.0168)
Observations	37,856	37,703	16,756	16,493
R-squared	0.03	0.03	0.03	0.03
Additional Controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2014 exchange rate shock, covering the period from November 18, 2014 to February 10, 2015. The dependent variable takes the value 1 if a depositor withdraws at least 50% of their total deposit balance during the shock period, and 0 otherwise Column (1) for each currency uses the full sample of banks. Column (2) excludes the a bank for which within-contract balance adjustments are unavailable. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table D2. Regression results for the 2020 Geopolitical Shock without two banks for which within-contract balance adjustments are unavailable

	AMD		USD	
	(1)	(2)	(1)	(2)
Insured Balance	0.0002***	0.0002***	0.1890***	0.1710***
	(0.0000)	(0.0000)	(0.0477)	(0.0541)
Active Saver	0.0188***	0.0142***	0.0266***	0.0152***
	(0.0018)	(0.0019)	(0.0030)	(0.0035)
Relationship Length	-0.0005**	-0.0006**	-0.0009*	-0.0007
	(0.0002)	(0.0002)	(0.0005)	(0.0005)
Number of Accounts	0.0001	0.0001	0.0001	0.0000
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Interest Rate	-0.0147***	-0.0160***	-0.0223***	-0.0258***
	(0.0009)	(0.0009)	(0.0016)	(0.0018)
Days to Maturity	0.0109***	0.0116***	0.0167***	0.0216***
	(0.0005)	(0.0006)	(0.0012)	(0.0014)
Insider	0.0075	0.0076	-0.0062	0.0024
	(0.0070)	(0.0071)	(0.0113)	(0.0123)
Above Insurance	0.0267***	0.0261***	0.0393***	0.0392***
	(0.0028)	(0.0031)	(0.0038)	(0.0043)
Inflow History	0.0194*	0.0213*	-0.0001	-0.0003
	(0.0103)	(0.0110)	(0.0006)	(0.0006)
Outflow History	0.0064**	0.0075***	0.0000***	0.0000***
	(0.0030)	(0.0027)	(0.0000)	(0.0000)
Observations	56,348	46,879	22,743	17,688
R-squared	0.02	0.02	0.04	0.04
Demographic Controls	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes

Source: Authors' estimations.

Note: This table reports the results of a regression examining the probability of a withdrawal during the 2020 geopolitical shock, covering the period from September 27, 2020 to December 31, 2020. The dependent variable takes the value 1 if a depositor withdraws at least 50% of their total deposit balance during the shock period, and 0 otherwise. Column (1) for each currency uses the full sample of banks. Column (2) excludes the two banks for which within-contract balance adjustments are unavailable. The unit of observation is individual–currency. All specifications control for demographic characteristics, including age, gender, and region as well as bank fixed effects. Heteroskedasticity-robust standard errors are shown in parentheses.

Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.